

Low Rank Matrix Approximation

Given a matrix $A \in \mathbb{C}^{m \times n}$ of rank r , find $B \in \mathbb{C}^{m \times n}$ which solves:

$$\min_B \|A - B\|_2$$

$$\text{s.t. } \underline{\text{rank}(B) = k} \quad (k \leq r)$$

Non-convex optimization problem

It admits closed-form solution:

SOLUTION:

Base case: $k = r$, $B^* = A$

General case: $k < r$. $A = U \begin{bmatrix} \sigma_1 & & 0 \\ & \ddots & \\ 0 & & \sigma_r & 0 \\ 0 & & 0 & 0 \end{bmatrix} V^H$
 $\Rightarrow \sigma_{k+1} > 0$

$$A = U \begin{bmatrix} \sigma_1 & \sigma_2 & \dots & \sigma_k & \sigma_{k+1} & \dots \\ & & & & & \\ & & & & & \end{bmatrix} \begin{bmatrix} \underbrace{V_{k+1}^H}_{k+1} \\ \underbrace{\tilde{V}^H}_{n-k-1} \end{bmatrix}$$

Consider any rank- k $B \dots (m \times n)$

Consider the matrix $C = BV_{k+1}$

$$\text{rank}(C) \leq k$$

Null space of C has dimension ≥ 1

There exists non-zero vector $z \in N(BV_{k+1})$

$$\Rightarrow \underbrace{BV_{k+1}} \cdot z = 0$$

Define:

$$\tilde{z} = \frac{V_{k+1} z}{\|V_{k+1} z\|_2}$$

Is $\tilde{z} \neq 0$?

Yes, since
 $\text{rank}(V_{k+1}) = k+1$

and $z \neq 0$

$$\Rightarrow \|A-B\|_2 \geq \frac{\|(A-B)\tilde{z}\|_2}{\|\tilde{z}\|_2}$$

$$= \|(A-B)\tilde{z}\|_2$$

$$= \|A\tilde{z} - B\tilde{z}\|_2 = \|A\tilde{z}\|_2$$

$$= \left\| U \begin{bmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_{k+1} \end{bmatrix} \begin{bmatrix} V_{k+1}^H \\ \tilde{z}^H \end{bmatrix} \frac{V_{k+1} z}{\|V_{k+1} z\|_2} \right\|_2$$

$$= \left\| U \begin{bmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_{k+1} \end{bmatrix} \begin{bmatrix} \frac{z}{\|V_{k+1} z\|_2} \\ 0 \end{bmatrix} \right\|_2$$

$$= \left\| U \begin{bmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_{k+1} \end{bmatrix} \begin{bmatrix} \frac{z}{\|V_{k+1} z\|_2} \\ 0 \end{bmatrix} \right\|_2$$

$$= \left\| \begin{bmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_{k+1} & \\ & & & \ddots \\ & & & & 0 \end{bmatrix} \begin{bmatrix} \frac{z}{\|z\|_2} \\ \\ 0 \end{bmatrix} \right\|_2$$

$$= \frac{1}{\|z\|_2} \sqrt{\sum_{i=1}^{k+1} \sigma_i^2 |z_i|^2}$$

$$\geq \frac{1}{\|z\|_2} \sigma_{k+1} \underbrace{\sqrt{\sum_{i=1}^{k+1} |z_i|^2}}_{\|z\|_2}$$

$$= \sigma_{k+1}$$

$$\Rightarrow \|A - B\|_2 \geq \sigma_{k+1} \quad \text{for any rank } k \text{ } B.$$

$$\|Uv\|_2^2$$

$$= v^H U^H U v$$

$$= v^H v$$

$$= \|v\|_2^2$$

Similarly

$$\|V_{k+1} z\|_2^2$$

$$= z^H V_{k+1}^H V_{k+1} z$$

$$= \|z\|_2^2$$

Convince yourself that

$$B^* = U \begin{bmatrix} \sigma_1 & & & \\ & \ddots & & \\ & & \sigma_k & \\ & & & 0 & \ddots & 0 \end{bmatrix} V^H$$

achieves

$$\|A - B^*\|_2 = \sigma_{k+1}$$

$$B^* = U \begin{bmatrix} \sigma_1 & & & 0 \\ & \ddots & & \\ & & \sigma_k & \\ 0 & & & 0 \end{bmatrix} V^H$$

is best k -rank

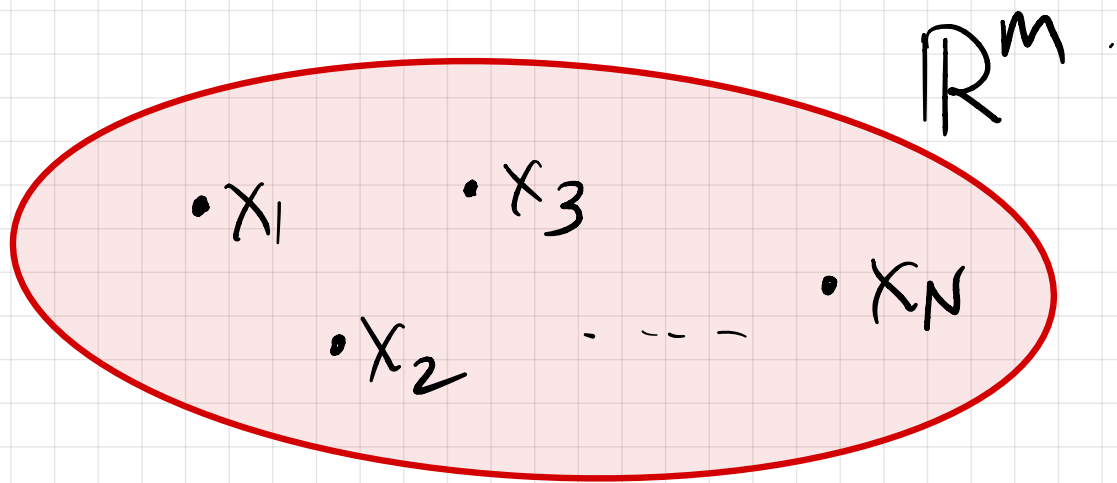
approx. of A .

PRINCIPAL COMPONENT ANALYSIS

"The Recipe"

Given N data points $x_1, x_2, \dots, x_N \in \mathbb{R}^M$, find the top K principal components (PC's).

(Assume we have subtracted the empirical mean from each data point)



RECIPE PROCEEDS AS:

1. COMPUTE EMPIRICAL COVARIANCE OF DATA

$$\Sigma = \sum_{i=1}^N x_i x_i^H = X X^H, \quad X = [x_1 x_2 \dots x_N]$$

2. COMPUTE SVD/EVD of Σ

$$\Sigma = U \begin{bmatrix} \sigma_1^2 & & 0 \\ & \sigma_2^2 & \\ 0 & & \sigma_M^2 \end{bmatrix} U^H$$

$$\sigma_1 \geq \sigma_2 \geq \sigma_3 \dots \geq \sigma_M$$

3. $U_1, U_2, \dots, U_K$ (singular vectors for top K singular values) ARE THE K "PC"s.

Most computationally intensive step is
computing SVD of Σ (size $M \times M$)

What if $M \gg N \geq k$?

One can follow the following "quicker" recipe:-

1. INSTEAD OF COMPUTING Σ , compute

$$\tilde{\Sigma} = X^H X \quad (\text{size } N \times N, \text{ much smaller than } M \times M)$$

2. COMPUTE SVD of $\tilde{\Sigma}$

$$\tilde{\Sigma} = V \begin{bmatrix} \sigma_1^2 & & 0 \\ & \ddots & \\ 0 & & \sigma_N^2 \end{bmatrix} V^H \quad (\text{same non zero singular values as } \Sigma)$$

Since $N \geq k$, collect top " k " singular vectors

$$v_1, v_2, \dots, v_k.$$

3. Reconstruct $u_1, u_2, \dots, u_k$ as

$$u_1 = \frac{1}{\sigma_1} A v_1, \quad u_2 = \frac{1}{\sigma_2} A v_2, \quad \dots, \quad u_k = \frac{1}{\sigma_k} A v_k$$

Verify these relations yourself.

Nice Recipes, but what exactly are we doing here?

What is the geometric interpretation of the "K" PC's $u_1, u_2, \dots, u_K$?

Answer: We are learning the "best" subspace of dim. K, that fits the data $x_1, x_2, \dots, x_N$.

BEST SUBSPACE via PCA

Given data points $x_1, x_2, \dots, x_N \in \mathbb{R}^M$, what is the best subspace of dim. K that fits the data?

Suppose $S \subset \mathbb{R}^M$ is a subspace of dimension K .

What is the projection error of projecting each point $x_1, x_2, \dots, x_N$ onto S ?

$$\text{Error of projecting } x_1 = \|x_1 - P_S x_1\|_2$$

$$\vdots$$
$$\text{Error of projecting } x_N = \|x_N - P_S x_N\|_2$$

[Here P_S is the projection matrix onto S]

The best subspace is the one where the sum of squares of the projection errors of all points are minimized.

This yields the following problem

$$\min_{S \subset \mathbb{R}^m} \sum_{i=1}^N \|x_i - P_S x_i\|_2^2 \quad \text{--- (P)}$$

such that $\dim(S) = K$.

Let us try to solve this problem when $K=1$.

Let S be a 1-dimensional subspace.

Let $\bar{v}$ be a unit-norm basis of S , i.e.

$$S = \text{span}\{\bar{v}\}, \quad \|\bar{v}\|_2 = 1.$$

Finding S is equivalent to finding $\bar{v}$.

$\Rightarrow$ Problem (P) reduces to

$$\min_{\bar{v} \in \mathbb{R}^m} \sum_{i=1}^N \|x_i - P_{R(\bar{v})} x_i\|_2^2 \quad \text{--- (P1)}$$

such that $\|\bar{v}\|_2 = 1$

Now $\sum_{i=1}^N \|x_i - P_{R(\bar{v})} x_i\|_2^2 = \sum_{i=1}^N \left(\|x_i\|_2^2 - \|P_{R(\bar{v})} x_i\|_2^2 \right)$

Due to orthogonality principle

$$\Rightarrow \min_{\bar{v}} \sum_{i=1}^N \|x_i - P_{R(\bar{v})} x_i\|_2^2 \stackrel{\text{is equivalent to}}{=} \max_{\bar{v}} \sum_{i=1}^N \|P_{R(\bar{v})} x_i\|_2^2$$

s.t. $\|\bar{v}\|_2 = 1$ s.t. $\|\bar{v}\|_2 = 1$

$$\equiv \max_{\bar{v}} \sum_{i=1}^N \|\bar{v} \bar{v}^H x_i\|_2^2, \text{ s.t. } \|\bar{v}\|_2 = 1$$

$$= \max_{\bar{v}} \sum_{i=1}^N |\bar{v}^H x_i|^2 \|\bar{v}\|_2^2, \text{ s.t. } \|\bar{v}\|_2 = 1$$

$$= \max_{\bar{v}} \sum_{i=1}^N |\bar{v}^H x_i|^2, \text{ s.t. } \|\bar{v}\|_2 = 1$$

$$= \max_{\bar{v}} \bar{v}^H \left(\sum_{i=1}^N x_i x_i^H \right) \bar{v}, \text{ s.t. } \|\bar{v}\|_2 = 1$$

$$= \max_{\bar{v}} \bar{v}^H X X^H \bar{v}, \text{ s.t. } \|\bar{v}\|_2 = 1$$

Solution is given by Rayleigh Quotient formulation.

$\bar{v}$ is an eigenvector corresponding to largest eigenvalue of $X X^H$, which is λ_1 .

The above result can be generalized to $k > 1$ dimensional subspaces by considering

$$S = \text{span} \{ \bar{v}_1, \bar{v}_2, \dots, \bar{v}_k \},$$

$$\text{when } \bar{V}^H \bar{V} = I. \quad (\bar{V} = [\bar{v}_1, \bar{v}_2, \dots, \bar{v}_k])$$

(i.e. - by finding an orthonormal basis of S).

$\bar{V}$ will comprise of $u_1, u_2, \dots, u_k$ as its columns.

APPLICATION AS CLASSIFIER

Given N faces, $x_1, x_2, \dots, x_N \in \mathbb{R}^M$, want to distinguish "happy" from "Neutral" faces.

- Step 1: Parse the data to separate the data according to their labels (happy or neutral).

Let $x_1, x_2, \dots, x_N$ represent Happy.
and $x_{N+1}, \dots, x_N$ " neutral.

- Step 2: Learn the "best subspace" that fits happy datapoints. (call it S_{happy})
Learn the "best subspace" that fits neutral datapoints (call it $S_{neutral}$)

